Everyday spreadsheet work

A first set of 20 user stories with observable acceptance criteria.

AUDIENCE AND STATUS

For a person working in the desktop spreadsheet interface. No SQL or Python is required to follow these stories. This is a draft backlog for review, not a claim of implemented features or an approved release scope.

SCOPE

Create, enter, calculate, organize, correct, save, and share. Configured workbook code participates in normal managed edits. Advanced SQL, code authoring, external APIs, and business workflows belong in later story sets.

Story group	IDs	Page
Create and open	01-02	2
Import and save	03-04	3
Navigate and edit	05-06	4
Copy and fill	07-08	5
Calculate	09-10	6
Organize the workbook	11-12	7
Format and read	13-14	8
Sort and filter	15-16	9
Find and correct	17-18	10
Share and validate	19-20	11

HOW TO USE

Each story states a user goal, four acceptance criteria, and a concrete example. IDs are stable for discussion. Criteria are proposed behavior; unresolved product choices are listed on page 12.

01 / Create and open

Begin with a blank workbook or return to existing work.

DH-US-01

Create a workbook

As a spreadsheet user, I want to start a blank workbook so I can capture information immediately.

ACCEPTANCE CRITERIA

1. Creating a workbook opens an editable sheet with a selected cell and visible row and column headings.

2. I can enter data without defining a database schema, writing code, or configuring a workflow.

3. The workbook and active sheet have visible names, and I can choose a native file location when saving.

4. Creating a new workbook leaves any existing open workbook and its stored content unchanged.

Example: Create a workbook, enter Item in A1, and enter Notebook in A2.

DH-US-02

Open an existing workbook

As a spreadsheet user, I want to reopen my workbook so I can continue where I left off.

ACCEPTANCE CRITERIA

1. Opening a valid native file restores its sheets, literal values, formula source, and saved formatting.

2. Calculated values are refreshed or visibly marked as pending; unverified cached values are not shown as current.

3. A missing, unreadable, or incompatible file produces an actionable error without overwriting the file.

4. If I only have read access, that state is visible and edits cannot silently appear to have been saved.

Example: Reopen a saved workbook and confirm that a formula remains editable as a formula.

02 / Import and save

Reuse existing information and keep a reliable native copy.

DH-US-03

Bring in an existing spreadsheet or CSV

As a spreadsheet user, I want to import existing files so I can reuse information I already have.

ACCEPTANCE CRITERIA

1. For supported CSV, Excel, and OpenDocument files, I can inspect the proposed sheets, columns, and conversions.

2. I can preserve an identifier such as 00123 as text; ambiguous dates and failed conversions are reported.

3. Unsupported formulas, formatting, and other features are identified before I accept a lossy conversion.

4. Applying the import reports its outcome and transaction scope; imported code cannot activate itself.

Example: Import a CSV with an ID column and preserve 00123 instead of converting it to 123.

DH-US-04

Save work and resume it reliably

As a spreadsheet user, I want to know my work is stored so I can close the application with confidence.

ACCEPTANCE CRITERIA

1. The interface distinguishes stored changes from edits or calculations that are still pending or have failed.

2. After a successful native save and reopen, values, formulas, sheets, and formatting match the stored state.

3. A save failure explains the problem and preserves any uncommitted edits that can still be retried.

4. Closing with pending edits offers a clear way to keep them or explicitly discard them; stored work is not lost.

Example: Save, close, and reopen a sheet containing values, a total formula, and number formatting.

03 / Navigate and edit

Work directly in the grid with familiar controls.

DH-US-05

Navigate and select cells

As a spreadsheet user, I want to move around and select ranges quickly so I can work efficiently.

ACCEPTANCE CRITERIA

1. Mouse and keyboard navigation show the active cell address and clearly distinguish the selected range.

2. I can select one cell, a rectangular range, a complete row, or a complete column.

3. I can jump to a valid address, including a distant cell, without manually scrolling through intervening rows.

4. Scrolling, selecting, and fetching visible cells do not change stored values or trigger data-edit actions.

Example: Jump to D5000, select D5000:F5010, and return to A1 without changing any data.

DH-US-06

Enter, edit, and clear values

As a spreadsheet user, I want to change cell content directly so I can maintain my information.

ACCEPTANCE CRITERIA

1. I can enter text, numbers, dates, and formulas using the supported value and locale rules.

2. Editing exposes the underlying value or formula; cancelling an edit restores the previous cell content.

3. I can intentionally enter numeric-looking text such as 00123 without losing its leading zeros.

4. Clear contents removes selected values or formulas while retaining formatting; clearing formatting is distinct.

Example: Change 12 to 15, cancel a second edit, then clear the value while keeping its number format.

04 / Copy and fill

Reuse values, layouts, and calculations across ranges.

DH-US-07

Copy, cut, and paste a range

As a spreadsheet user, I want to move and reuse ranges so I can avoid entering the same information twice.

ACCEPTANCE CRITERIA

1. Copying and pasting a rectangle preserves its row and column layout, with explicit values-only paste available.

2. Internal formula copies follow the reference rules in DH-US-08, and normal paste preserves supported formats.

3. Cut content is removed from its source only if the destination change succeeds as part of the same operation.

4. Tab-separated clipboard text maps to cells; a rejected ordinary paste leaves source and destination unchanged.

Example: Copy a 3-by-2 range to a new location, then paste its calculated results as values.

DH-US-08

Fill values and formulas

As a spreadsheet user, I want to fill a range from a pattern so I can extend repeated calculations quickly.

ACCEPTANCE CRITERIA

1. Filling a constant repeats it across the selected destination range without modifying unrelated cells.

2. Relative references move with each copied formula; absolute row or column components remain fixed.

3. An explicit numeric-series fill continues a selected pattern; the destination range is visible before applying it.

4. A failed fill is reported and does not leave a partly applied ordinary fill operation.

Example: Fill =C2*B1 from E2 to E3; E3 contains =C3*B1.

05 / Calculate

Build formulas and understand the state of their results.

DH-US-09

Calculate with familiar formulas

As a spreadsheet user, I want to use formulas so I can calculate results from the values in my workbook.

ACCEPTANCE CRITERIA

1. I can enter arithmetic expressions and functions in the declared first compatibility tier, including SUM and IF.

2. The grid shows a calculated result while the formula entry area retains the editable formula source.

3. A range expression such as =SUM(D2:D4) evaluates the referenced values under documented type rules.

4. An invalid or unsupported expression shows a meaningful error instead of silently becoming zero or plain data.

Example: With B2 = 2 and C2 = 3.50, =B2*C2 returns 7; SUM can total several result cells.

DH-US-10

See updated results and understand errors

As a spreadsheet user, I want dependent results to update so I can trust what the sheet currently shows.

ACCEPTANCE CRITERIA

1. After an input changes, dependent results update or clearly indicate that calculation is still pending.

2. A completed calculation from an older input state cannot overwrite a result for a newer edit.

3. A calculation error identifies the affected cell and cause; circular references are visibly identified.

4. Correcting the cause allows dependent results to recover; a displayed error does not destroy the formula.

Example: Change B2 from 2 to 3 and see =B2*C2 change from 7 to 10.50 when C2 is 3.50.

06 / Organize the workbook

Change rows, columns, and sheets without losing meaning.

DH-US-11

Insert and remove rows or columns

As a spreadsheet user, I want to change the sheet structure so I can make room for new information.

ACCEPTANCE CRITERIA

1. I can insert or delete selected rows or columns, and content outside the affected structure is preserved.

2. An inserted row shifts existing content and updates positional references to the same surviving cells.

3. Deleting a directly referenced cell produces a visible broken-reference result rather than pointing elsewhere.

4. If a structural operation fails validation, its data changes and related reference changes are not partly applied.

Example: Insert a row above row 3; a reference to the original A3 follows that cell to A4.

DH-US-12

Organize work across sheets

As a spreadsheet user, I want to manage multiple sheets so I can separate related parts of my work.

ACCEPTANCE CRITERIA

1. I can add, rename, reorder, and duplicate a sheet; each resulting sheet has a distinct valid name.

2. Switching sheets leaves other sheets intact, and a duplicate initially preserves supported content and formats.

3. Cross-sheet formulas can reference another sheet, and renaming that sheet preserves the intended reference.

4. Deleting a referenced sheet makes the broken reference visible and never silently redirects it to another sheet.

Example: Rename Inputs to Source; a formula referencing Inputs!A1 still uses the same cell.

07 / Format and read

Make information readable while retaining its stored meaning.

DH-US-13

Format cells without changing their meaning

As a spreadsheet user, I want to format values and headings so my sheet is easy to read and interpret.

ACCEPTANCE CRITERIA

1. I can apply supported text emphasis, alignment, fills, borders, and number formats to a selected range.

2. Number formatting changes display without replacing the underlying stored value or formula.

3. Clearing formatting retains cell content, while copying formats applies presentation to the destination.

4. Native save and reopen preserve supported formatting and the distinction between displayed and stored values.

Example: Display 0.125 as 12.5% and 3.14159 as 3.14 while retaining their original numeric values.

DH-US-14

Keep a large sheet readable

As a spreadsheet user, I want to adjust the view so I can understand the data while moving around.

ACCEPTANCE CRITERIA

1. I can resize rows and columns, wrap text, and fit a column to its visible content.

2. Freezing header rows or key columns keeps them visible while the rest of the sheet scrolls.

3. I can hide and unhide rows or columns, with a visible indication that hidden content exists.

4. Layout changes retain stored values and formula definitions, and the saved view is restored on reopen.

Example: Freeze the header row, widen the description column, and scroll without losing the column labels.

08 / Sort and filter

Inspect lists while keeping complete records intact.

DH-US-15

Sort related rows together

As a spreadsheet user, I want to sort a list so I can inspect its records in a useful order.

ACCEPTANCE CRITERIA

1. The sort scope and header treatment are explicit before applying the selected ascending or descending order.

2. Sorting a table or declared list moves complete records together, preserving the relationships between columns.

3. Sorting by multiple keys uses the declared key precedence and documented blank and mixed-type ordering.

4. Stored record identities remain stable; failed sorts leave the original data order and values intact.

Example: Sort IDs and labels by label; each ID stays paired with its original label.

DH-US-16

Filter a list without deleting data

As a spreadsheet user, I want to show matching rows so I can focus on the information I need.

ACCEPTANCE CRITERIA

1. I can apply supported value and comparison filters, and the active criteria are visible.

2. Filtering changes which rows are shown without deleting nonmatching rows or changing their stored values.

3. Clearing filters restores all rows, and the interface distinguishes a filtered result from an empty dataset.

4. An edit in a filtered view targets the intended underlying record even if that edit changes whether it matches.

Example: Filter Category to Office, edit one visible description, then clear the filter and see all rows.

09 / Find and correct

Locate repeated content and recover from ordinary mistakes.

DH-US-17

Find and replace content

As a spreadsheet user, I want to locate and replace content so I can correct repeated information efficiently.

ACCEPTANCE CRITERIA

1. I can choose a selection, active sheet, or workbook as the search scope and move between matching cells.

2. Search options distinguish displayed values from formula source, and matches show their locations.

3. Before Replace All, I can inspect the scope and match count; a single replacement changes only its target.

4. Ordinary Replace All is one managed operation; a validation failure leaves the affected cells unchanged.

Example: Find a repeated label within the active sheet and replace it without editing another sheet.

DH-US-18

Undo and redo ordinary edits

As a spreadsheet user, I want to reverse mistakes so I can explore and edit without losing control of my work.

ACCEPTANCE CRITERIA

1. For supported actions, undo restores prior values, formulas, or formatting and refreshes dependent results.

2. A paste or fill is one understandable undo action; redo reapplies it when no intervening edit invalidates redo.

3. Undo and redo participate in applicable workbook rules; rejection leaves the current state intact.

4. The interface explains an unavailable or unsupported undo action and does not claim to reverse external side effects.

Example: Paste into six cells, undo once to restore those cells, and redo to reapply the paste.

10 / Share and validate

Exchange useful copies and understand configured workbook behavior.

DH-US-19

Export a usable copy

As a spreadsheet user, I want to export my work so I can share it with people using other spreadsheet tools.

ACCEPTANCE CRITERIA

1. I can select the supported CSV, Excel, or OpenDocument format and the sheet or data scope to export.

2. The export uses a consistent stored state with current calculated values, or clearly reports pending calculations.

3. Before accepting a lossy export, I see unsupported features, including Python code with no executable equivalent.

4. A failed or oversized export is reported without silent truncation, and exporting leaves the native workbook intact.

Example: Export a supported sheet to Excel and its values to CSV, then verify totals in both outputs.

DH-US-20

Understand workbook rules during normal editing

As a spreadsheet user, I want workbook rules to run consistently so I can make valid edits without writing code.

ACCEPTANCE CRITERIA

1. An ordinary edit or paste runs applicable configured Python handlers automatically through the application.

2. If a transactional handler rejects an edit, the original change and its generated edits are rolled back together.

3. After a successful edit, I see accepted values, handler-generated changes, and current or pending calculation state.

4. An error explains the affected operation; committed data with a failed follow-up is clearly distinguished from rollback.

Example: Paste several labels when a handler disallows blanks; one blank rejects the ordinary paste as a whole.

A typical session

Use this small fixture to walk through the first story set from entry to export.

Item	Quantity	Unit price	Amount
Notebook	2	3.50	=B2*C2 -> 7.00
Pen	4	1.25	=B3*C3 -> 5.00
Folder	3	2.00	=B4*C4 -> 6.00

1. Create and enter. Create a workbook, name a sheet Supplies, and enter the headers and three rows shown above. (01, 06, 12)

2. Calculate and format. Enter =B2*C2 in D2 and fill down to D4. Enter =SUM(D2:D4) in D6; the total is 18.00. Format prices and amounts to two decimals. (08, 09, 13)

3. Update and inspect. Change B3 from 4 to 6. D3 becomes 7.50 and D6 becomes 20.50. Filter Quantity to values above 2, then clear the filter. (10, 16)

4. Correct and share. Copy and undo a range edit, save, reopen, and export through a supported format. Confirm the native formulas remain editable and the exported values match. (02, 04, 07, 18, 19)

DECISIONS TO REFINE

Persistence: explicit Save, autosave, or both, and the meaning of pending edits.

References: exact sort, duplicate-sheet, insertion, and deletion adjustment rules.

Compatibility: first function/format tiers, numeric and date conversion rules.

Undo: supported actions, history lifetime, and handler behavior during reversal.

Responsiveness: reference datasets and measurable navigation and calculation targets.

DESIGN BASIS AND ACCESS BOUNDARY

Based on AGENTS.md and docs/design/README.md, data-model.md, python-runtime.md, and interfaces-and-interoperability.md. These stories propose everyday behavior while leaving implementation choices open.

GUI edits use the managed engine and applicable Python hooks. Independent SQLite edits bypass those hooks; reopening does not replay skipped edit handlers. This boundary is unchanged by the story set.